

Internship Projects:

Labeling UI

Objective: Develop a lightweight, mobile-responsive web application that enables analysts to label multi-modal data stored in a shared directory (e.g., Google Drive/S3) to support **Multi-Modal** and **Entity-level Sentiment Analysis**.

1. Core System Requirements

- **Data Source Integration:** The UI must read a central "Source CSV" and serve images/text to the user.
- **State Management:** The system must track the "Last Labeled Index" so multiple users don't label the same row and work is never lost.
- **Mobile-First Design:** Optimized for "swipe-style" or quick-tap labeling so researchers can label data on the go.
- **Multi-tasks:** enable multiple task options for different labeling tasks. A user can define and choose labeling tasks.
- **Output:** Every session must append results to a "Master Labels" file or database.

2. Specialized Labeling Workflows

Workflow A: Image-Text Relationship (Multi-modal)

- **Input:** Image + Accompanying Text.
- **Label Options:**
 1. **Independent:** The image alone explains the topic.
 2. **Context-Dependent:** The image adds value but requires the text to make sense.
 3. **Noise:** The image is irrelevant to the text.

Workflow B: Semantic & Emotion (Entity-Centric)

- **Input:** Text snippet.
- **Label Steps:**
 1. **Entity Identification:** Highlight or type the main entity (Person, Org, Place).
 2. **Topic Assignment:** Write the text topic.
 3. **Sentiment/Emotion:** Rate the framing (e.g., Good, Bad, Trust, Fear, Anger) specifically toward that entity.

Workflow C: Caption

- **Feature:** A text box for the user to write a **"Golden Caption"** describing the image.

Drone oversight

Task 1: The "Footprint" Calculator

Goal: Create a function that calculates the geographical "bounding box" of a single image on the ground.

- **Input:** Drone GPS (lat, lon), Altitude, Camera Gimbal Angle, and Field of View (FOV) of the camera.
- **Outcome:** The four GPS coordinates of the image corners on a flat map.
- **Mentor Tip:** Start with a "Nadir" (straight down) view and then graduate to "Oblique" (angled) views.

Task 2: The "Heatmap" Coverage Grid

Goal: Build a grid-based system to track which areas have been scanned and which are "blind spots."

- **Input:** The "Footprint" coordinates from Task 1.
- **Outcome:** A 2D grid where cells change color based on coverage, where we can filter by drone IDs.
 - **Gray:** Not scanned.
 - **Green:** Scanned once.
 - **Red:** Overlap (scanned by multiple drones/images).
- **Mentor Tip:**
 - Ensures that Drone A's data doesn't get mixed up with Drone B's data.
 - This is a visual task. Use libraries like **Leaflet.js** or **Folium** so one can see the results on a real map.

Task 3: Image "Stitching" & Mosaicking

Goal: Build a pipeline to align and merge multiple overlapping images into a single large map (Orthomosaic).

- **Input:** A set of overlapping drone images with their telemetry, can be the footprints or the heatmap.
- **Outcome:** A single, large, georeferenced image file.

Task 4: Image-to-GPS Projection

Goal: If a user clicks a pixel in an image, the system should estimate the GPS coordinate of that point.

- **Input:** Image pixel (x, y) + Drone telemetry.

- **Outcome:** A point from the 2D image onto the 3D world map.

Task 5: Altitude-Adaptive Scaling

Goal: Create a "Flight Parameter Calculator" that adjusts speed and altitude based on the required Ground Sample Distance (GSD/Resolution).

- **Input:** Desired image resolution (e.g., 2cm/pixel) and camera sensor specs.
- **Outcome:** A recommendation for altitude and speed."
- **Mentor Tip:** Need to understand the relationship between altitude and resolution.

Task 6: Mission Planner (Area-to-Waypoint Logic)

Goal: Develop an algorithm that takes a polygon (Area of Interest) and generates a flight path (waypoints) for a drone.

- **Input:** Map coordinates of an area, desired overlap percentage, altitude, speed, and drone model specs (Field of view).
- **Outcome:** A list of GPS waypoints and altitudes that ensure 100% coverage of the area.